

Percent Application: Receipts - Project

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Objectives:

- (1.) I will be able to calculate tax and total price given the initial price and the tax rate.
- (2.) I will be able to calculate tax rate when given the subtotal, tax amount, shipping, and total.
- (3.) I will be able to calculate the tax and tax rate when given the subtotal and total price
- (4.) I will be able to find the exact percentage of a discount when given rounded versions and the remaining information on a receipt.
- (5.) I will be able to find the tax rate of two different orders and explain why they are different even though they were bought from the same supplier and had the same shipping address.

Case 1

Given: initial price, tax rate

To Calculate: tax, total price



Let Men's Socks = Item

(a.)

$$5.3\% = \frac{5.3}{100} = 0.053$$

Item

Cost = 12.48

$$Tax = 0.053(12.48)$$

$$= 0.66144$$

$$\approx \$0.66$$

(b.)

$$Total\ Price = 12.48 + 0.66144$$

$$= 13.14144$$

$$\approx \$13.14$$



Case 2

Note: Shipping is not included in the calculations of tax rate.

Given: subtotal, shipping, tax amount, and total.

To calculate: Tax Rate

 Hide order summary ^		\$72.08
	The Black Parade (CD)	\$13.98
	Bullets Track Longsleeve T-Shirt X Large	\$40.00
	Revenge Pencil Set	\$4.99
Subtotal		\$58.97
Shipping		\$9.99
Taxes		\$3.12
Total		USD \$72.08

Subtotal = \$58.97

$$Tax = \$3.12$$

$$Total = 72.08 - 9.99 = \$62.09$$

What percent of \$58.97 is \$3.12?

$$\frac{\text{is}}{\text{of}} = \frac{\text{what}\%}{100}$$

$$\frac{3.12}{58.97} = \frac{\text{what}\%}{100}$$

$$\text{what} \times 58.97 = 3.12 \times 100$$

$$\text{what} = \frac{3.12 \times 100}{58.97}$$

$$\text{what} = 5.290825844\%$$

Case 3

Given: Tax Rate and Tax

Calculate:

(a.) Initial Price (Subtotal)

(b.) Total Price



WAL*MART
TBD Mgr. TBD
ANNISTON, AL
ST# 329 OP# 9035 TE# 35 TR# 397

SKLS ATL SL	026254300000	F	
0.99 lb @	1.0 lb /8.41		
SKLS ATL SL	026254300000	F	
1.02 lb @	1.0 lb /8.41		
SKLS ATL SL	026254300000	F	
0.91 lb @	1.0 lb /8.41		
BAG GARLIC	007096900247	F	
SUBTOTAL			
TAX	2 9 %		2.65
TOTAL			
VISA DEBIT TEND			
VISA	**** *	3802	
CHANGE DUE			0.00

ITEMS SOLD 4

TC# 3049 6934 8749 2967 0764



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Let the subtotal (sum of the initial prices) of the price of the items = p

(a.)

$$9\% = \frac{9}{100} = 0.09$$

Item

$$\textit{Subtotal} = p$$

$$\textit{Tax} = 0.09(p) = 0.09p$$

$$\textit{Tax} = 2.65$$

$$\Rightarrow$$

$$0.09p = 2.65$$

$$p = \frac{2.65}{0.09}$$

$$p = 29.\overline{44}$$

$$\textit{Subtotal} \approx \$29.44$$

(b.)

$$\textit{Total Price} = 29.\overline{44} + 2.65$$

$$= \$32.09\overline{44}$$

$$\approx \$32.09$$



WAL*MART
TBD Mgr. TBD
ANNISTON, AL
ST# 329 OP# 9035 TE# 35 TR# 397

SKLS ATL SL	026254300000 F	
0.99 lb @ 1.0 lb /8.41		8.33
SKLS ATL SL	026254300000 F	
1.02 lb @ 1.0 lb /8.41		8.58
SKLS ATL SL	026254300000 F	
0.91 lb @ 1.0 lb /8.41		7.65
BAG GARLIC	007096900247 F	4.88

SUBTOTAL		29.44
TAX 2	9 %	2.65
TOTAL		32.09
VISA DEBIT TEND		32.09
VISA **** * 3802		
CHANGE DUE		0.00

ITEMS SOLD 4

TC# 3049 6934 8749 2967 0764



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Case 4

Given: initial price, discount, sales price.

Calculate the exact percent discount.

 ETSY INC September 28, 2024 . Payment		-\$4.20
Paid with	Seller info	
Bank of America Customized Cash Rewards Visa (VISA Credit Card [REDACTED]) You'll see "PAYPAL *ETSY INC" on your card statement. on September 28, 2024	\$4.20 ETSY INC [REDACTED] no-reply@etsy.com	
	Purchase details	
	Celestial Black Phone Icons Pack I Android and Iphone Icons Item #1242188242	\$14.00
Ship to [REDACTED] [REDACTED] [REDACTED] United States	Transaction Discoun	- \$9.80
	Total	\$4.20

Initial Price = \$14.00

Let % Discount = p

$$p\% = \frac{p}{100} = 0.01p$$

$$\text{Discount} = p\% \text{ of } 14.00$$

$$= 0.01p(14.00)$$

$$= 0.14p$$

$$\Rightarrow$$

$$0.14p = 9.80$$

$$p = \frac{9.80}{0.14}$$

$$p = 70$$

$$\therefore \% \text{ discount} = 70\%$$

Check

$$70\% \text{ of } \$14$$

$$0.7 \times \$14 = \$9.80$$

Case 5

Ryan's mom ordered two pieces of clothing from Amazon on two separate days.

They were both sent to the same address in Virginia.

Neither had food or drinks.

However, they were charge different tax rates. Find the tax rate for each.

1st Order:

Order Summary

Items:	\$21.99
Shipping & Handling:	\$8.00
Total Before Tax:	\$29.99
Estimated Tax Collected:	\$1.59
Order Total	\$31.58

2nd Order

Order Summary

Items:	\$88.62
Shipping & Handling:	\$0.00
Total Before Tax:	\$88.62
Estimated Tax Collected:	\$4.69
Order Total	\$93.31

To Calculate:

- (a.) Calculate the tax rate of the 1st order.
- (b.) Calculate the tax rate of the 2nd order.
- (c.) Explain why this happens.

Let the subtotal (sum of the initial prices) of all the items = p

(a.)

1st Order

What % of 21.99 is 1.59?

Percent – Equation

$$\frac{\text{what}}{100} \times 21.99 = 1.59$$

$$\text{what} = \frac{1.59 \times 100}{21.99}$$

$$\text{what} = 7.230559345$$

$$\text{Tax rate} = 7.230559345\%$$

(b.)

2nd Order

What % of 88.62 is 4.69?

Percent – Proportion

$$\frac{\text{is}}{\text{of}} = \frac{\text{what}\%}{100}$$

$$\frac{4.69}{88.62} = \frac{\text{what}\%}{100}$$

$$\text{what}\% = \frac{4.69 \times 100}{88.62}$$

$$\text{what}\% = 5.292259084\%$$

$$\text{Tax rate} = 5.292259084\%$$

(c.)

The reason that Amazon has different taxes depending on the order or day, even if they're going to the same address and paid with the same card, is because it is an international seller with different sellers in different countries. So, when calculating the regional shipping taxes and other taxes that come along with running a business, Amazon must combine all of that to create a specific tax amount for the product because even if they're all coming from Amazon, one product's local tax laws can be different from another's. For example, if one package is being produced in France, and the other in Italy, Amazon's job is to take the sales price of said items and apply all of the overlapping international taxes to create the final price for the consumer.

Works Cited

- “Arithmetic/Algebra Project: Percents: Show Me Your Receipts.” Samuel Chukwuemeka. *MTH 154: Quantitative Reasoning (Onsite)*, <https://conferencepresentations.appspot.com/Projects/ArithmeticAlgebra/Receipts.html>. Accessed 24 October 2024.
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